Executive Summary

Carrier Performance in Windy Departure Conditions at JFK

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Purpose

This report continues our review of operations at John F. Kennedy International Airport by examining how weather affects departures across the carriers that fly out of JFK. Specifically, this analysis answers the question: which carrier has the highest percentage of its flights departing in windy conditions, defined here as a recorded wind speed greater than 20 miles per hour at the time of departure? Understanding this helps identify which carriers are most frequently exposed to a weather condition associated with delays, turbulence, and operational disruption.

Key Finding

Endeavor Air Inc. (carrier code 9E) has the highest percentage of windy departures among the sixteen carriers examined, at 8.31%. Of its 17,266 flights in the dataset, 1,434 departed when wind speed exceeded 20 mph. JetBlue Airways follows closely at 7.37%, and Hawaiian Airlines at 7.06%, though Hawaiian's result is based on a much smaller number of flights (340) and should be read with that in mind. The full results, ranked from highest to lowest percentage, are shown below.

Carrier	Code	Total Flights	Windy Departures	% Windy
Endeavor Air Inc.	9E	17,266	1,434	8.31%
JetBlue Airways	B6	53,704	3,958	7.37%
Hawaiian Airlines Inc.	HA	340	24	7.06%
American Airlines Inc.	AA	31,917	2,180	6.83%
Delta Air Lines Inc.	DL	47,530	3,240	6.82%
Envoy Air	MQ	24,963	1,669	6.69%
Virgin America	VX	5,099	332	6.51%
AirTran Airways Corporation	FL	3,166	201	6.35%
Mesa Airlines Inc.	YV	541	31	5.73%
US Airways Inc.	US	19,781	1,110	5.61%
Frontier Airlines Inc.	F9	679	35	5.15%
Southwest Airlines Co.	WN	12,014	573	4.77%
United Air Lines Inc.	UA	57,621	2,572	4.46%
ExpressJet Airlines Inc.	EV	50,969	2,130	4.18%

Carrier	Code	Total Flights	Windy Departures	% Windy
Alaska Airlines Inc.	AS	707	17	2.40%
SkyWest Airlines Inc.	OO	29	0	0.00%

Note: Carriers with very low flight volumes in this dataset — SkyWest (29 flights) and Hawaiian Airlines (340 flights) — are included for completeness but should be interpreted with caution, since a percentage based on so few flights is far more sensitive to chance than one based on tens of thousands of flights.

Methodology

The flight data stores departure and arrival times as plain numbers (for example, 1408 for 2:08 PM) rather than as true timestamps, and it does not directly record the weather at the moment each flight left the gate. To answer the manager's question, this analysis used a multi-step SQL query built with Common Table Expressions (CTEs), which let us break the problem into clear, sequential stages rather than one dense query:

1. Convert the raw date and time fields into a proper timestamp for each flight's scheduled departure, so departure time could be compared against the hourly weather record.
2. Match each flight to the weather recorded at its origin airport during the same year, month, day, and hour as its departure.
3. Flag each flight as a “windy departure” if the matched wind speed exceeded 20 mph, then count, for each carrier, how many of its total flights fell into that category.
4. Calculate each carrier's percentage of windy departures and rank carriers from highest to lowest.

The two queries below carry out this process. The first illustrates the timestamp conversion; the second builds on that logic to produce the final ranked results shown in the table above.

Query 1: Building Departure and Arrival Timestamps

```
WITH flight_times AS (
    SELECT
        f.*,
        (f.dep_time / 100) AS dep_hour,
        make_timestamp(
            f.year, f.month, f.day,
            (f.dep_time / 100), (f.dep_time % 100), 0
        ) AS dep_timestamp,
        make_timestamp(
            f.year, f.month, f.day,
            (f.arr_time / 100), (f.arr_time % 100), 0
        ) AS arr_timestamp
    FROM jfk.flights f
    WHERE f.dep_time IS NOT NULL
      AND f.arr_time IS NOT NULL
      AND f.dep_time BETWEEN 0 AND 2359
      AND f.arr_time BETWEEN 0 AND 2359
)
SELECT carrier, flight, dep_timestamp, arr_timestamp
FROM flight_times;
```

Query 2: Percentage of Windy Departures by Carrier

```
WITH flight_times AS (
    SELECT
        f.*,
        (f.dep_time / 100) AS dep_hour,
        make_timestamp(
            f.year, f.month, f.day,
            (f.dep_time / 100), (f.dep_time % 100), 0
        ) AS dep_timestamp
    FROM jfk.flights f
    WHERE f.dep_time IS NOT NULL
      AND f.dep_time BETWEEN 0 AND 2359
),
flight_weather AS (
    SELECT
        ft.carrier, ft.flight, ft.dep_timestamp, w.wind_speed
    FROM flight_times ft
    JOIN jfk.weather w
      ON w.origin = ft.origin
     AND w.year   = ft.year
     AND w.month  = ft.month
     AND w.day    = ft.day
     AND w.hour   = ft.dep_hour
),
carrier_windy AS (
    SELECT
        carrier,
        COUNT(*) AS total_flights,
        COUNT(*) FILTER (WHERE wind_speed > 20) AS windy_flights
    FROM flight_weather
    GROUP BY carrier
)
SELECT
    a.c2 AS carrier_name,
    cw.carrier,
    cw.total_flights,
    cw.windy_flights,
    ROUND(100.0 * cw.windy_flights / cw.total_flights, 2) AS pct_windy_departures
FROM carrier_windy cw
JOIN jfk.airlines a ON a.c1 = cw.carrier
ORDER BY pct_windy_departures DESC;
```

Is This a Good Metric?

Percentage of windy departures is a reasonable and defensible metric for comparing carriers, for three main reasons:

1. It accounts for differences in carrier size. Raw counts of windy flights would automatically favor larger carriers like United (57,621 total flights) over smaller ones like Hawaiian (340 total flights). Using a percentage puts every carrier on the same footing, regardless of how many flights it operates.

2. It is based on an objective, repeatable threshold. Wind speed above 20 mph is a clear, measurable condition rather than a subjective judgment call, so the same analysis will produce the same results if repeated or audited by someone else.
3. It provides an early, actionable signal. Carriers or route planners with a consistently higher percentage of windy departures may benefit from closer review of scheduling patterns, de-icing or ground-delay procedures, or contingency planning, making this a useful starting point for operational conversations rather than an end point.

Limitations of This Metric

At the same time, this metric has real limitations that should be kept in mind before drawing firm conclusions:

1. Windy departure does not necessarily mean a problem occurred. A high percentage only shows exposure to wind, not whether that wind actually caused a delay, a rough takeoff, or any other negative outcome. A carrier could have a high percentage simply because it operates more flights during naturally windier parts of the day or year.
2. The 20 mph threshold is a simplification. It treats a flight departing in 21 mph wind the same as one departing in 50 mph wind, and it does not account for wind gusts or wind direction relative to the runway, both of which matter more to pilots than sustained wind speed alone.
3. The weather match is approximate. Weather is recorded once per hour at the origin airport, while flights depart at specific minutes within that hour, so the wind speed used may be up to nearly 60 minutes removed from the actual moment of departure. Weather at the destination or along the flight path is not considered at all.

What Additional Data Would Improve This Metric

A more complete picture would benefit from a few additional data points. Actual arrival and departure delay figures, matched against windy versus calm departures, would let us test whether wind speed is actually associated with worse outcomes, rather than simply measuring exposure. Wind gust speed and wind direction relative to each runway would offer a more operationally meaningful measure than sustained wind speed alone, since crosswinds affect takeoff and landing differently than headwinds or tailwinds. Finally, incorporating weather at the destination airport and along the flight route, not just at JFK, would give a fuller sense of the conditions each flight actually experienced from gate to gate.

Conclusion

Based on the data available, Endeavor Air Inc. experiences the highest percentage of windy departures among carriers operating out of JFK. This finding is useful as a starting point for further investigation, but should be paired with actual delay and outcome data before being used to draw conclusions about carrier performance or reliability.